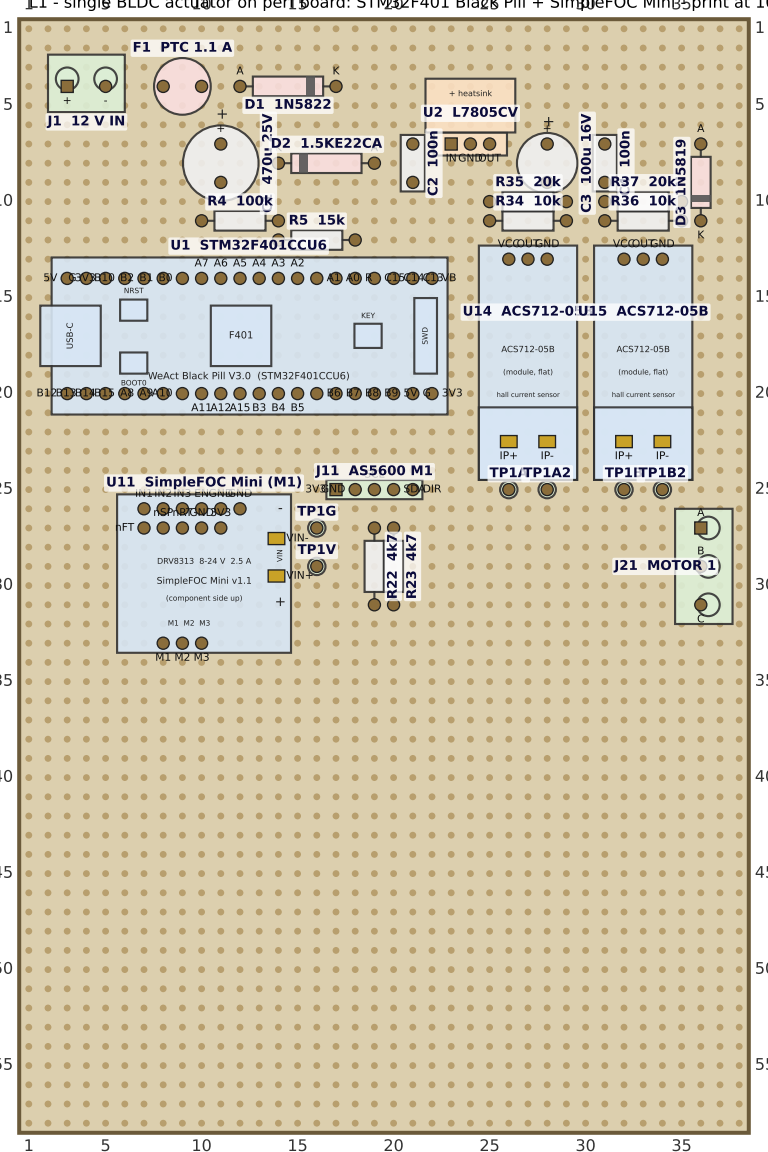

11 - single BLDC actuator on perf board: STM32F401 Black Pill + SimpleFOC Mini print at 10
F1 PTC 1.1 A
J1 12 V IN
D1 1N5822
D2 1.5KE22CA
470u 25V
R4 100k
R5 15k
U1 STM32F401CCU6
U2 L7805CV
+ heatsink
IN GND OUT
C2 100n
R35 20k
R34 10k
C3 100u 16V
R37 20k
R36 10k
C4 100n
D3 1N5819
U14 ACS712-05B
U15 ACS712-05B
ACS712-05B
(module, flat)
hall current sensor
VCC OUT GND
IP+ IP-
TP1 TP1A2
TP1 TP1B2
WeAct Black Pill V3.0 (STM32F401CCU6)
USB-C
F401
SWD
KEY
NRST
BOOT0
U11 SimpleFOC Mini (M1)
IN1 IN2 IN3 EN GND VM
SCL SDA GND 3V3
nFT
DRV8313 8-24 V 2.5 A
SimpleFOC Mini v1.1
(component side up)
M1 M2 M3
VIN
TP1G
TP1V
R22 4k7
R23 4k7
J11 AS5600 M1
3V3 DIR
J21 MOTOR 1